An Accurate Semi-Lagrangian Scheme for Raindrop Sedimentation

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ABSTRACT

An accurate method for explicitly computing the transport of rainwater is devised and tested. The scheme is conservative and positive definite. A cubic polynomial is employed to approximate the spatial distribution of the rainwater; thus, the resulting scheme is of high accuracy. With a slope modification, numerical oscillations can be easily eliminated in the vicinity of large gradients. Being essentially of Lagrangian type, the scheme is stable even with a Courant–Friedrichs–Lewy number larger than 1. The scheme was tested with the numerical example of Kato and was found to have a computational stability similar to the box-Lagrangian scheme, but it is more accurate.

1. Introduction

The motion of raindrops plays an important role in the phenomena related to precipitation. This is not only because of its direct role in bringing about rainfall but also because of its significant impact on the surrounding atmosphere. For example, the drag effect of raindrops can be of great importance in precipitation-dominant events (Kato and Saito 1995). Thus, meteorological simulations where the fine structures of precipitation systems are to be resolved should explicitly include the calculation of the rainwater sedimentation.

Since the typical fall speed of raindrops is on the order of several meters per second, the requirement for computational stability in the vertical direction will restrict the time integration interval if any Eulerian explicit transport scheme is used. This will apparently affect the computational efficiency of the whole numerical model, especially in the case where fine vertical resolution is required for the planetary boundary layer. A time-splitting, which uses a finer time integration interval with an Eulerian scheme for the precipitation sedimentation, can make the computation stable. However, reducing the time integration interval means an increase in the step-number of time marching and consequently a larger accumulation in numerical errors. Being an alternative solution, an algorithm which is stable even for a Courant–Friedrichs–Lewy (CFL) number larger than 1 is worth exploring.

In order to get rid of the stability restriction on conventional Eulerian methods, numerical methods following Lagrangian thought have been suggested. Grabowski and Smolarkiewicz (1996) devised a semi-Lagrangian method to the rain-evolution equation in an anelastic cloud model. Their comparisons show that the semi-Lagrangian approach is more efficient than the traditional Eulerian approach. However, the semi-Lagrangian scheme used in Grabowski and Smolarkiewicz (1996) is based on the nonconservative form, which does not assure the mass conservation of the rainwater. Kato (1995) suggested a so-called box-Lagrangian scheme for the motion of raindrops. Kato's formulation introduces a measure of volume, that is, the so-called box, which leads to an exact conservative approximation. Using a Lagrangian invariant mapping, the box-Lagrangian scheme appears free of the restriction of computational stability in terms of CFL number. However, in Kato's box-Lagrangian scheme, a piecewise

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constant approximation is used to estimate the spatial distribution of the rainwater. This, as mentioned by Kato (1995) himself, results in a first-order scheme. A well-known drawback of the first-order scheme is its excessive numerical diffusion. Fine structures of the transported quantity will be smeared out if a first-order scheme is used. Thus, a scheme of higher order is desirable for situations where it is of interest to resolve small-scale variations.

The semi-Lagrangian schemes were devised and have shared popularity in atmospheric modeling. A major merit of semi-Lagrangian schemes is that they can be easily designed to be stable even under a large CFL number. Although how to get an accurate solution under a large CFL number for a drastically changing velocity field still remains an unsolved problem, less time integration makes the semi-Lagrangian scheme efficient. A traditional semi-Lagrangian scheme is designed to solve the first-order hyperbolic equation of advection (or convection) form, which state a Lagrangian invariant solution along the trajectory of the characteristic velocity field. When the departure point for a Lagrangian particle that arrives at a grid point is located among the surrounding computational grid points where the field variable is defined, interpolation approximation needs to be constructed. So far, many studies in the literature have been devoted to how to construct interpolation functions. Among the widely used are cubic Lagrangian interpolation (McDonald 1984, 1987) and cubic spline (Purnell 1976; Riishøjgaad et al. 1998). Some high-order interpolation functions of good quality, such as monotonicity and shape-preserving, have also been explored (Rasch and Williamson 1990; Holnicki 1995; Bermejo and Conde 2002). Some applications, comprehensive evaluations, and surveys of semi-Lagrangian schemes can be found in Williamson et al. (1998), Bartello and Thomas (1996), and Staniforth and Côté (1991). Another type of interpolation construction is known as the cubic interpolated propagation (CIP) method (Yabe and Aoki 1991; Xiao 2000), in which the first-order spatial derivatives of the interpolation function are assumed to follow a set of governing equations derived by differentiating the advection equation with respect to the spatial coordinates. By the special treatment of the first-order derivatives of the interpolation function, the CIP reconstruction uses only one mesh cell to construct the interpolation profile and gives superior numerical dispersions.

However, different from most Eulerian methods, which are derived by using control volume or a cell-integrated average, the Lagrangian invariant solution used in a conventional semi-Lagrangian scheme is based only on a single spatial point and is usually cast in a nonconservation form. Few studies on the conservation of the semi-Lagrangian schemes have been reported so far in the traditional school of semi-Lagrangian thought because of the intrinsic difficulties in introducing a conservation formulation in a semi-Lagrangian scheme. In fact, conservative Eulerian schemes of flux form can be extended as a kind of conservative semi-Lagrangian scheme, which can be computationally stable for a large CFL number. An example is the flux-form semi-Lagrangian (FFSL) scheme of Lin and Rood (1996). Yabe et al. (2001) suggested a semi-Lagrangian method that conserves exactly the transported physical field. Xiao and Yabe (2001) introduced the slope at the cell center as another constraint on the interpolation function and devised a class of conservative and oscillationless schemes.

In this paper, we present a high-order computational scheme for rainwater based on the interpolation reconstruction of Xiao and Yabe (2001). Similar to the box-Lagrangian scheme of Kato (1995), the present scheme is computationally stable even if the CFL is larger than unity. The present method has a remarkably higher accuracy than the box-Lagrangian scheme regarding both the reconstruction of the distribution profile and the estimation of trajectory. Section 2 describes the numerical scheme. A few remarks on the algorithmic differences among the box-Lagrangian scheme, the FFSL scheme of Lin and Rood (1996), and the present scheme will be also included. Numerical results will be given in section 3. The benchmark test of Kato (1995) is used to evaluate the present method as a numerical scheme for rainwater transport, and the paper ends with a short summary in section 4.

2. Numerical scheme

The transport of rainwater in the vertical direction can be mathematically expressed by

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial z}(wf) = 0, \tag{1}$$

where t refers to the time and z is the vertical coordinate. The quantity f is the rain content (ρq), ρ is air density, and q is rain mixing ratio.

The advection velocity w is the composition of the velocity of the background air flow and the fall speed of the raindrops. The latter is usually determined through an expression related to the amount of rainwater. In the tests described in this paper, we consider only the effect of the rain fallout and adopt the relationship for the terminal fall speed of rain (Kato 1995) as

$$w(z, t) = \kappa_1[f(z, t)]^{\kappa_2}. \tag{2}$$

In the calculations of the present paper the constants are defined as $\kappa_1 = -12.1115$ and $\kappa_2 = 0.125$ (Kessler 1969). We can find some other formulations expressing the terminal fall speed in terms of rainwater mass, such as that used in Klemp and Wilhelmson (1978), but they do not look essentially different from (2); thus, in the present paper we will use (2) as an adequate relation to describe the nonlinearity between the fall speed and the rainwater mass. As can be expected, the nonlinearity of the equation will create shocklike compression waves

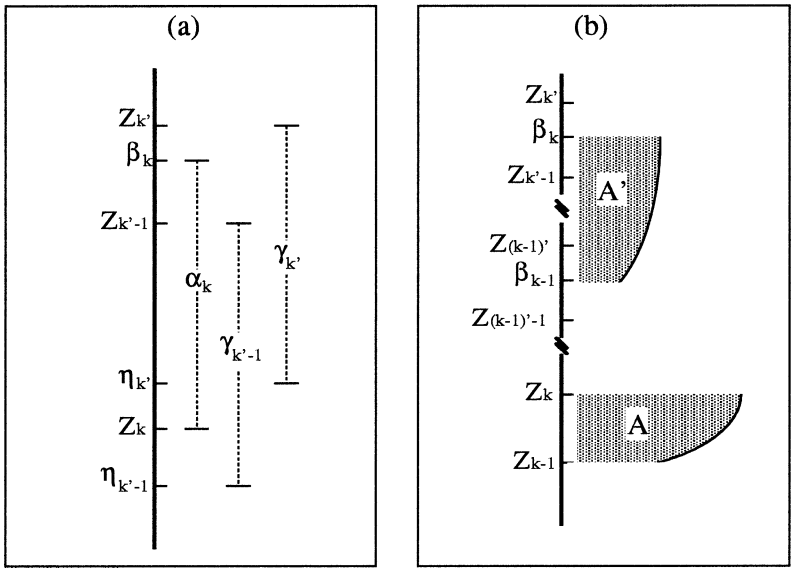

FIG. 1. Schematic illustration of the numerical scheme. (a) The departure point β_k for grid point Z_k is effectively determined by a forward-searching procedure, that is, first compute the displacement ($\gamma_{k'}$) and the arrival point ($\eta_{k'}$) based on the advection velocity defined by (5), then calculate the departure point β_k for grid point Z_k via (6), (8), and (9), which make use of the results from the previous forward-searching step. (b) A schematic of how to remap the mass onto grid cell (Z_{k-1}, Z_k). After the departure points (β_k and β_{k-1}) are determined for Z_k and Z_{k-1}, the cell-integrated value within (Z_{k-1}, Z_k) at a new time step can be computed by integrating the cubic polynomial over (β_{k-1}, β_k) as in (20). In this figure and the context, $\eta_{k'-1}$ and $\eta_{k'}$ denote the arrival points that satisfy $\eta_{k'-1} < Z_k < \eta_{k'}$, and $Z_{k'-1}$ and $Z_{k'}$ are the grid points that satisfy $Z_{k'-1} < \beta_k < Z_{k'}$.

and rarefaction waves and thus increases the discontinuities in the water mass and the fall speed of the rain.

Equation (1) can be also written in an advection form as

$$\frac{\partial f}{\partial t} + w\frac{\partial f}{\partial z} = -f\frac{\partial w}{\partial z}. \tag{3}$$

Equations (1) and (2) or (3) can be solved with properly posed initial conditions and boundary conditions.

The computing grid is defined as a set of discrete fixed points along the z axis as $Z_k(k \in \mathcal{K} = \{0, 1, 2, \ldots k\text{max}\})$, and $Z_{k_1} < Z_{k_2}$ for any $k_1 < k_2$. As in any semi-Lagrangian scheme, the numerical procedure of the present scheme involves both the estimation of the departure point and the construction of the interpolation of f. They will be respectively discussed as follows.

a. Calculation of the departure points

In the context of any semi-Lagrangian scheme, the solution procedure is simply based on the understanding that the Lagrangian invariant solution exists for pure advection on the trajectory path that connects departure point and arrival point. The determination of the departure point for any computational grid point is of great importance, and some schemes for that purpose have been devised (Temperton and Staniforth 1987; McGregor 1993). When a velocity field has large jumps or discontinuities, determining the departure point using the velocity of current time step at arrival point, which is usually called backward searching, will cause significant errors. In the rain sedimentation model, the fall speed depends on the water mass at the departure point; thus a so-called forward-searching approach should be more appropriate.

Let us consider the departure point (denoted by β_k in Fig. 1), by which we mean the starting point of a Lagrangian particle that arrives at a computational grid point, say Z_k. We start with a forward-searching step and compute the displacements of all particles using the velocities defined at their starting points, that is, the grid points. We introduce another set of subscript index $_{k'}$ and denote the computational grid points also as $Z_{k'}(k \in \mathcal{K} = \{0, 1, 2, \ldots k\text{max}\})$. As shown in Fig. 1a, given the velocity at step $n(t = t^n)$ for all grid points ($w^n_{k'}$ for all $k' \in \mathcal{K}$), we first compute the arrival point $\eta_{k'}$ for $Z_{k'}$ by

$$\eta_{k'} = Z_{k'} + \gamma_{k'} \quad \text{for all } k' \in \mathcal{K}, \tag{4}$$

where $\gamma_{k'}$ denotes the distance that a Lagrangian particle departing from $Z_{k'}$ travels during one time step Δt.

In any Lagrangian scheme or semi-Lagrangian scheme, a ''frozen'' velocity has to be always assumed during each time step. In the present scheme, $\gamma_{k'}$ is calculated by the advection velocity w defined in (2) as

$$\gamma_{k'} = \kappa_1(f^n_{k'})^{\kappa_2}\Delta t. \tag{5}$$

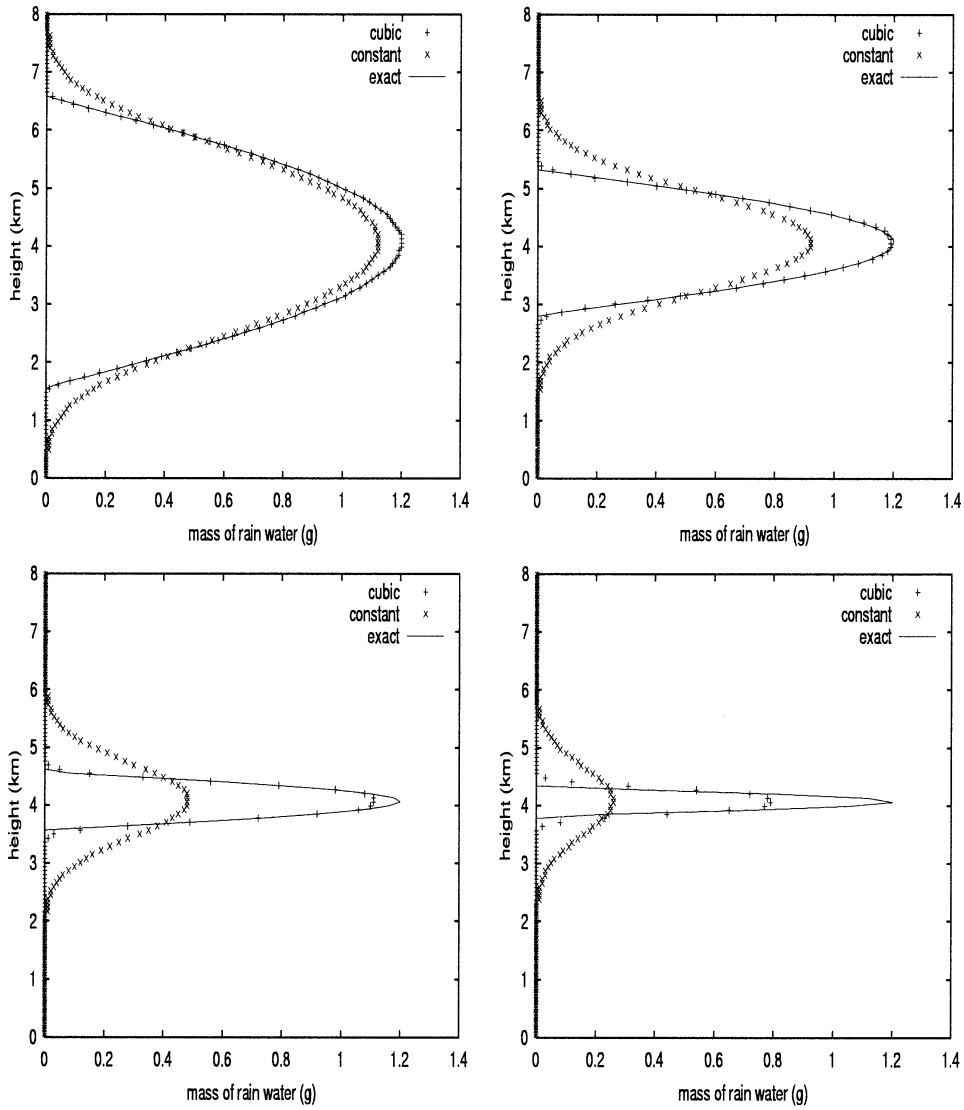

FIG. 2. Linear translation of cosine curves with different scales. The scale is defined in terms of the half-width of each cosine function as (top left) $D_z = 37.7\Delta z$, (top right) $D_z = 17.8\Delta z$, (bottom left) $D_z = 7.1\Delta z$, and (bottom right) $D_z = 3.6\Delta z$. Numerical results of both the present scheme (denoted as cubic) and the first-order scheme (equivalent to that of Kato and denoted as constant) are given. Plotted are the mass of the rainwater in a volume of 1 m^3 at $t = 20$ min.

After $\eta_{k'}$ are obtained for all grid point $k' \in \mathcal{K}$, the averaged effective advection velocity $\overline{w}_k$ for grid point Z_k is simply computed by using a linear interpolation based on the two neighboring arrival points $\eta_{k'-1}$ and $\eta_{k'}$ that satisfy $\eta_{k'-1} < Z_k < \eta_{k'}$:[1]

$$\overline{w}_k = \frac{1}{\eta_{k'} - \eta_{k'-1}} \frac{[\overline{\gamma}_{k'-1}(\eta_{k'} - Z_k) + \gamma_{k'}(Z_k - \eta_{k'-1})]}{\Delta t}. \tag{6}$$

In order to reduce the undulations caused by the large jump in the displacements of two neighboring trajectories, $\tilde{\gamma}_{k'-1}$ in (6) is modified as

$$\overline{\gamma}_{k'-1} = \begin{cases} \gamma_{k'-1}, & \text{if } |\eta_{k'} - \eta_{k'-1}| > \lambda_1(Z_k - Z_{k-1}) \\ \lambda_2 \gamma_{k'} + (1 - \lambda_2)\gamma_{k'-1} & \\ \quad \text{otherwise.} & \end{cases} \tag{7}$$

In the particular problem of raindrops, $\lambda_1 = 0.22$ and

[1] A more general formulation can be written as

$$\overline{w}_k = \frac{1}{\eta_{k'} - \eta_{\hat{k}'}}[\overline{\gamma}_{\hat{k}'}(\eta_{k'} - Z_k) + \gamma_{k'}(Z_k - \eta_{\hat{k}'})]/\Delta t,$$

where $\eta_{\hat{k}'}$ is the largest value of η_j with $j \le k' - 1$ and $\eta_j < Z_k < \eta_{k'}$.

$\lambda_2 = 0.95$ are used. This serves effectively as an ''artificial diffusion'' to reduce the discontinuities in the velocity field that are generally intensified because of the nonlinearity. There is an arbitrariness in choosing λ_1 and λ_2, but we found that the values given work well with the numerical experiments.

The displacement of the particle that arrives at Z_k at time step $n + 1$ is estimated by a Taylor approximation up to third order (McGregor 1993) as

$$\alpha_k = -\Delta t \overline{w}_k + \frac{1}{2}(\Delta t)^2 \overline{w}_k \left(\frac{\partial \overline{w}}{\partial z}\right)_k - \frac{1}{6}(\Delta t)^3 \overline{w}_k \left[\left(\frac{\partial \overline{w}}{\partial z}\right)_k^2 + \overline{w}_k \left(\frac{\partial^2 \overline{w}}{\partial z^2}\right)_k\right], \quad (8)$$

where $(\partial \overline{w}/\partial z)_k$ and $(\partial^2 \overline{w}/\partial z^2)_k$ are computed by central differencing. The departure point for Z_k is obtained by

$$\beta_k = Z_k - \alpha_k. \quad (9)$$

b. Construction of the interpolation function

As discussed in the previous section, we assume that the departure point for grid point Z_k, denoted by β_k, locates within grid cell $(Z_{k'-1}, Z_{k'})$; that is, $Z_{k'-1} < \beta_k < Z_{k'}$.

As shown in Fig. 1b, an interpolation function needs to be constructed within $(Z_{k'-1}, Z_{k'})$. We use the piecewise cubic-interpolation reconstruction devised by Xiao and Yabe (2001) to approximate the distribution of the rainwater on each grid cell.

Considering the interpolation reconstruction on $(Z_{k'-1}, Z_{k'})$, we suppose that the integrated average value, the first-order derivative at cell center, and the cell-interface values at the two ends of the cell are given for time step n as $\overline{f}^n_{k'-1/2}$, $d^n_{k'-1/2}$, $f^n_{k'-1}$, and $f^n_{k'}$. The piecewise interpolation function can be written as

$$F_{k'-1/2}(z) = c_{0k'-1/2} + c_{1k'-1/2}(z - Z_{k'-1}) + c_{2k'-1/2}(z - Z_{k'-1})^2 + c_{3k'-1/2}(z - Z_{k'-1})^3, \quad \text{for } z \in (Z_{k'-1}, Z_{k'}). \quad (10)$$

From the given constraint conditions, we have

$$F_{k'-1/2}(Z_{k'-1}) = f^n_{k'-1}, \quad (11)$$

$$F_{k'-1/2}(Z_{k'}) = f^n_{k'}, \quad (12)$$

$$\frac{dF_{k'-1/2}}{dx}(Z_{k'-1/2}) = d^n_{k'-1/2}, \quad \text{and} \quad (13)$$

$$\frac{1}{\Delta z_{k'-1/2}} \int_{Z_{k'-1}}^{Z_{k'}} F_{k'-1/2}(z)\, dz = \overline{f}^n_{k'-1/2}, \quad (14)$$

where $\Delta z_{k'-1/2} = Z_{k'} - Z_{k'-1}$.

The cubic polynomial (10) can be completely determined from (11)–(14), and the coefficients read

$$\begin{aligned} c_{0k'-1/2} &= f^n_{k'-1}, \\ c_{1k'-1/2} &= \frac{6}{\Delta z_{k'-1/2}} \overline{f}^n_{k'-1/2} - \frac{6}{\Delta z_{k'-1/2}} f^n_{k'-1} - 2d^n_{k'-1/2}, \\ c_{2k'-1/2} &= -\frac{6}{\Delta z^2_{k'-1/2}} \overline{f}^n_{k'-1/2} + \frac{3}{\Delta z^2_{k'-1/2}}(3f^n_{k'-1} - f^n_{k'}) + \frac{6}{\Delta z_{k'-1/2}} d^n_{k'-1/2}, \\ c_{3k'-1/2} &= -\frac{4}{\Delta z^3_{k'-1/2}}(f^n_{k'-1} - f^n_{k'}) - \frac{4}{\Delta z^2_{k'-1/2}} d^n_{k'-1/2}. \end{aligned} \quad (15)$$

The first-order derivative of the interpolation function at the cell center, $d^n_{k'-1/2}$ is used to control the numerical oscillation. A slope limiter for positive definiteness can be simply devised by modifying the formulation suggested by van Leer (1979),

$$d^n_{k'-1/2} = \begin{cases} \dfrac{1}{\Delta z_{k'-1/2}} \operatorname{sign}(S^0_{k'-1/2}) \min(|S^0_{k'-1/2}|, |2S^+_{k'-1/2}|, |2S^-_{k'-1/2}|) & \text{if } S^+_{k'-1/2} \times S^-_{k'-1/2} > 0 \quad \text{or} \quad DS^0_{k'-1/2} \le 0 \\ 0 & \text{otherwise,} \end{cases} \quad (16)$$

where $DS^0_{k'-1/2} = (f^n_{k'+1/2} - 2f^n_{k'+1/2} + f^n_{k'-3/2})$, $S^0_{k'-1/2} = (f^n_{k'+1/2} - f^n_{k'-3/2})/2$, $S^+_{k'-1/2} = (f^n_{k'+1/2} - f^n_{k'-1/2})$ and $S^-_{k'-1/2} = (f^n_{k'-1/2} - f^n_{k'-3/2})$.

The value of the interpolation function at the cell center $Z_{k'-1/2}$ can be obtained from interpolation function (10) as

$$f^n_{k'-1/2} = \frac{3}{2}\overline{f}^n_{k'-1/2} - \frac{1}{4}(f^n_{k'-1} + f^n_{k'}). \quad (17)$$

c. Mapping the transported quantities

The final step of the numerical procedure is to map the interpolation profile from the departure cell(s) to the arrival cell.

As shown in Fig. 1b, providing the departure points corresponding to the two interfaces of cell (Z_{k-1}, Z_k) (i.e., β_{k-1} and β_k) fall in $[Z_{(k-1)'-1}, Z_{(k-1)'}]$ and $(Z_{k'-1}, Z_{k'})$, respectively; our task is to map the total rainwater between β_{k-1} and β_k, indicated by region A' in Fig. 1b,

at the nth step onto (Z_{k-1}, Z_k) for the $(n + 1)$th step. After the interpolation functions are constructed for all cells, the numerical solution of the interface value f is predicted from (3). Its value at time step $n + 1$ is first updated as a Lagrangian invariant, that is, the value of the piecewise interpolation function $F_{k'-1/2}$ at $(\beta_k - Z_{k'-1})$,

$$f_k^* = F_{k'-1/2}(\beta_k - Z_{k'-1}), \tag{18}$$

and corrected according to the variation of velocity field

$$f_k^{n+1} = f_k^*\left\{1 - \frac{1}{2}\left[\left(\frac{\partial w^*}{\partial z}\right)_k + \left(\frac{\partial w^n}{\partial z}\right)_{k'}\right]\Delta t\right\}. \tag{19}$$

The velocity divergence is calculated by averaging the values at departure point $(\partial w^n/\partial z)_{k'}$ and arrival point $(\partial w^*/\partial z)_k$, and w^* is obtained from (2) with f^* calculated by (18). In the present model, both $(\partial w^n/\partial z)_{k'}$ and $(\partial w^*/\partial z)_k$ are computed by a central difference approximation.

The cell-integrated average for cell (Z_{k-1}, Z_k) at the $(n + 1)$th step, $\overline{f}_{k-1/2}^{n+1}$ (shown as region A in Fig. 1b), is calculated by mapping the corresponding region (shown as A' in Fig. 1b) as

$$\overline{f}_{k-1/2}^{n+1} = \Delta z_{k-1/2}^{-1}\left[\int_{\beta_{k-1}}^{Z_{(k-1)'}} F_{(k-1)'-1/2}(z)\, dz + \sum_{j=(k-1)'+1}^{k'-1} \Delta z_{j-1/2}\overline{f}_{j-1/2}^{n} + \int_{Z_{k'-1}}^{\beta_{(k)}} F_{k'-1/2}(z)\, dz\right]. \tag{20}$$

In the above expression the integration is calculated by using the piecewise interpolation function discussed in the above section.

d. Remarks

It is easy to show that the piecewise constant approximation used in Kato (1995) is a degeneration of (10). A piecewise constant interpolation can be obtained by only letting $\overline{f}_{k'-1/2}^{n} = f_{k'-1}^{n} = f_{k'}^{n}$ in (10). As shown in Kato (1995), the box-Lagrangian scheme is equivalent to the first-order upwind Eulerian scheme when the CFL number is less than unity. Using a cubic interpolation, the present scheme can get an accuracy higher than third order.

The way of searching for the arrival point in Kato (1995) is actually the so-called forward searching. Our way is computing the departure point of the trajectory via an interpolation and can be called backward searching. However, in the present scheme, searching for the departure point is completely based on a ''preforward-searching'' procedure, that is, estimating $\gamma_{k'}$ via (5). Therefore, similar to the box-Lagrangian scheme, the present method removes the restriction on the time step

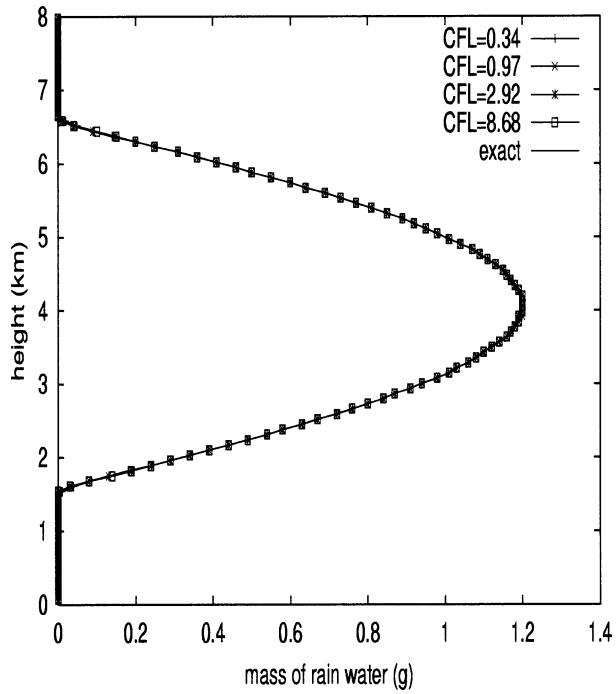

FIG. 3. Linear transport of a cosine bell with the half-width being $D_z = 37.7\Delta z$. Plotted are the mass of the rainwater in a volume of 1 m³ computed with different CFL numbers.

interval due to the computational stability in terms of CFL number. However, it is very difficult to find accurately the trajectory path for a large time step once there are drastic changes in velocity field, and a large time step may even lead to a computational instability. The practical CFL number in the present scheme is also subject to a bound for stability as shown later.

Mapping (20) looks similar to the flux-form semi-Lagrangian method of Lin and Rood (1996), in which any Eulerian scheme in a flux-form can be stabilized for large CFL calculations by extending the spatial-integration path for the flux calculations. However, there are some substantial differences between the two regarding the following aspects: 1) In the present scheme, we first compute the Lagrangian solutions as the point values for cell interfaces. It is called point mapping. The second mapping is called volume mapping, which transports the volume-integrated value of the departure region to the arrival region. In fact, a volume remapping is equivalent to a flux formulation; thus mass conservation is guaranteed. 2) Concerning the interpolation reconstruction, the present scheme uses both the point value and the cell-integrated average. Moreover, the present scheme gives a continuous interpolated profile over the whole computational domain, while the other conventional high-resolution Eulerian schemes, such as monotone upstream-centered scheme for conservation laws (MUSCL; van Leer 1979) and piecewise parabolic method (PPM; Colella and Woodward 1984), tend to

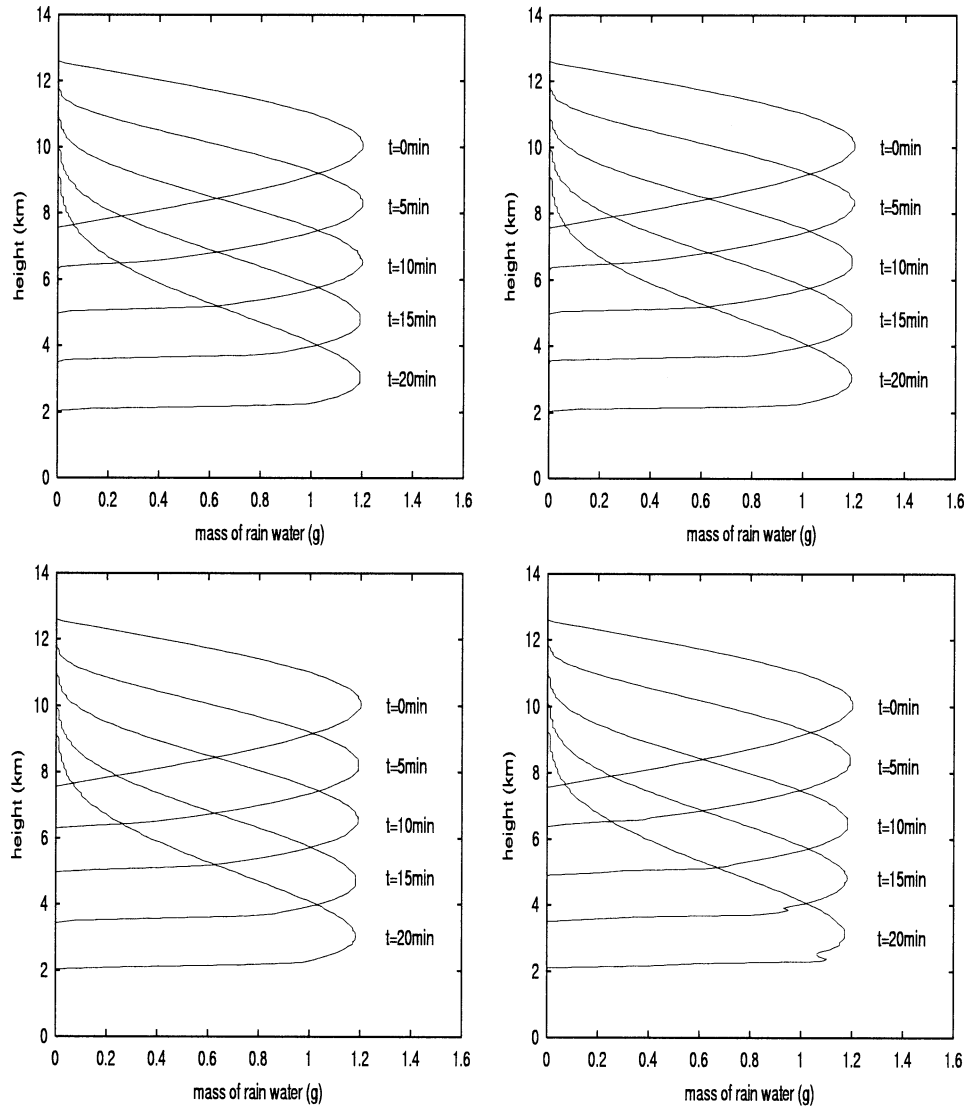

FIG. 4. Snapshots for the mass of rainwater per cubic meter. Displayed are the results with different time step intervals: (top left) $\Delta t = 6$ s, (top right) $\Delta t = 12$ s, (bottom left) $\Delta t = 24$ s, and (bottom right) $\Delta t = 36$ s.

create discontinuities across cell interfaces to enforce monotonicity.

Concerning the computational efficiency, the present scheme is more expensive than Kato's box-Lagrangian model and requires almost the same operations as other high-order schemes with monotone enforcement, like the PPM scheme (Colella and Woodward 1984), for the interpolation reconstruction and the mapping step. The searching for departure point is computed through a noniterative procedure and requires a computational effort similar to that in the box model of Kato (1995).

3. Numerical tests

The initial distribution of the rainwater is defined as (Kato 1995)

$$f^0(z) = \begin{cases} 1.2 \times \cos\left(\dfrac{\pi}{2} \times \dfrac{Z_c - z}{D_z}\right) & \text{for } |Z_c - z| \le D_z \\ 0 & \text{otherwise.} \end{cases} \tag{21}$$

The value f is expressed in grams per cubic meter and Z_c is fixed as $Z_c = 1.0 \times 10^4$ m. The computational domain is $0 \le z \le 1.4 \times 10^4$ m. An equally spaced grid with $\Delta z = 70$ m is used for all of the numerical experiments.

We first computed a linear translation of a cosine curve. A constant fall speed is specified as $w = -5$ m s^{-1} through the whole computational domain, and the time step interval is $\Delta t = 2.5$ s. In order to evaluate the capability of the present scheme in capturing the

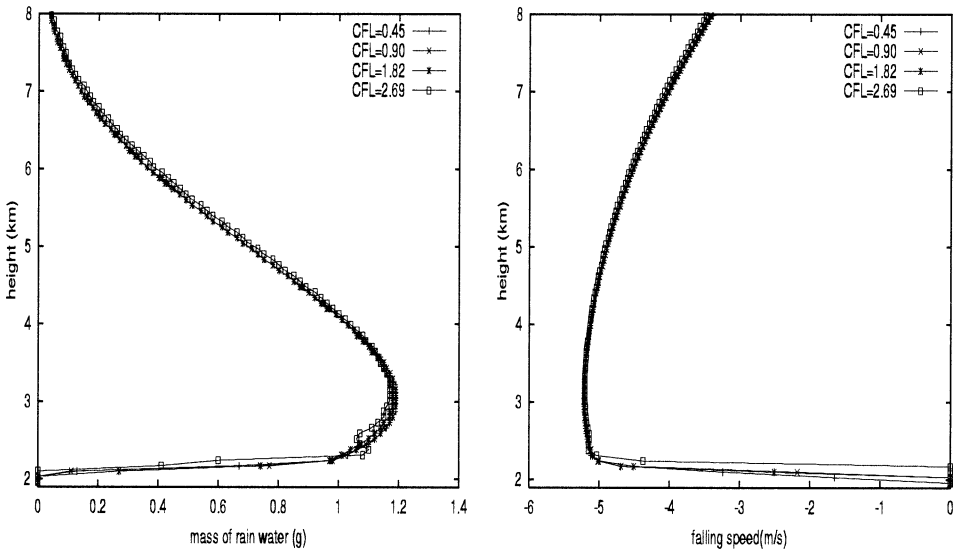

FIG. 5. Comparison of the numerical results at $t = 20$ min with different CFL numbers. Displayed are (left) the mass of rainwater per cubic meter and (right) the falling speed.

variations of different spatial scales, we computed cosine curves with different half-width D_z. For comparison with the piecewise constant approximation, which has been used in the box-Lagrangian method (Kato 1995), we also include the results based on such a low-order approximation. In this test the box-Lagrangian scheme is identical to the first-order upwind scheme.

Shown in Fig. 2, the first-order scheme, labeled as constant in the figure, is much more diffusive. With enough grid points or for a structure much larger than the grid spacing, the first-order scheme seems to be able to give reasonable results, but the numerical diffusion error becomes very significant for small scales. The peak value of the distribution is damped to less than one-half of the true solution as shown in the results of $D_z = 7.1\Delta z$ and $D_z = 3.6\Delta z$. This means that a first-order scheme will not be adequate to resolve a disturbance smaller than $14\Delta z$. If we assume that the typical grid spacing in the vertical direction for a free atmosphere is 100–200 m, a disturbance with a vertical scale of less than 2 km cannot be reasonably simulated by the first-order scheme.

Our scheme produced more accurate solutions to all scales when compared with the first-order scheme. Large-scale variations are almost exactly simulated by the present scheme. Even for the severe test of small-scale disturbance ($D_z = 3.6\Delta z$), the amplitude of the initial disturbance is still preserved to 70% whereas the first-order scheme to 20%.

For a uniform velocity field, the scheme is stable no matter how large the time-integration interval is. Figure 3 shows the numerical results of the present method for different CFL numbers with $w = -5$ m s^{-1}. The numerical results are almost identical to the exact solution, even for very different CFL numbers. We also computed disturbances of smaller spatial scales and find that the stability as shown in Fig. 3 does not depend on the initial disturbances in this type of passive transport problems. A larger-time-step calculation requires less integration steps and causes less accumulations in numerical diffusions, thus giving a better-preserved amplitude to short waves.

We tested the present scheme with the mass-dependent falling speed defined by (2). As in Kato (1995), $D_z = 2.5 \times 10^3$ m is used. The grid spacing is $\Delta z = 70$ m. We computed with different Δt between 6 and 36 s and gave the numerical results at different simulation times in Fig. 4. As can be expected, a shocklike front gradually develops because of the nonlinearity. In this test, the maximum falling speed is 5.22 m s^{-1}, and so the corresponding CFL numbers for the different Δt are 0.45, 0.90, 1.82, and 2.69, respectively. The computation is stable even for a CFL number larger than 1. Not much difference is observed among the results from different time-integration intervals. A further comparison is conducted by plotting the final results at $t = 20$ min for various time step intervals (Δt) in Fig. 5. The present method is insensitive to time-step interval and gives an accurate solution even with a large Δt.

When the advection velocity varies, the time-integration interval Δt should be bounded for both numerical accuracy and computational stability. A stability condition for semi-Lagrangian schemes in terms of the so-called Lipschitz number or deformational CFL (DeCFL) number has been discussed by Smolarkiewicz and Pudykiewicz (1992) and Lin and Rood (1996). The stability condition for a semi-Lagrangian scheme can be stated as

$$\mathrm{DeCFL} \equiv -\frac{(w_k^n - w_{k-1}^n)}{\Delta z_{k-1/2}}\Delta t \leq 1, \quad \text{for } k \in \mathcal{K}. \quad (22)$$

Inequality (22) is in fact of more physical significance

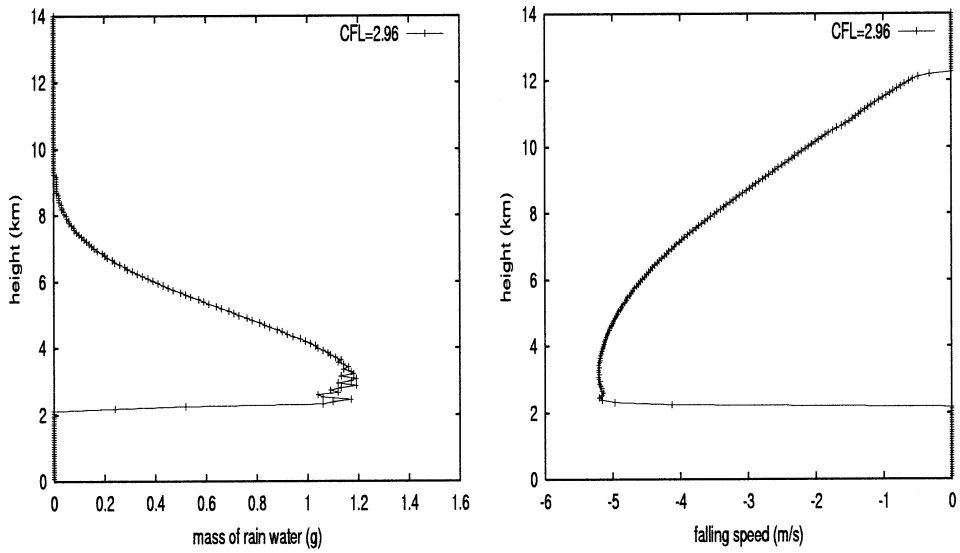

FIG. 6. (left) The mass of rainwater per cubic meter at $t = 20$ min with $\Delta t = 39.5$ s, and (right) corresponding falling speed.

than the numerical one. It means that any two trajectories cannot cross each other. The effective advection velocity computed by (6) gets around to some extent the restriction of the computational stability in respect to the velocity variation. The largest DeCFL number for the raindrop model is larger than one in the calculations with $\Delta t = 24$ s, but the solution is stable, as shown in Fig. 4, even after the shocklike front has fully developed.

As can be seen from Fig. 5, a small wiggle is observed right behind the front at the final instant in the results of $\Delta t = 36$ s. When the front gets steeper and the discontinuity in the velocity field becomes significant, it is very difficult to get accurate solutions with a large time integration interval. Figure 6 shows the numerical results for $\Delta t = 39.5$ s. The local instability becomes striking in the region behind the shock front. Our numerical tests show that the present scheme is stable if $|\mathrm{DeCFL}| < 2.4$. In the raindrop model tested in this paper, the permissible CFL number for computational stability is about 3, but, as shown in Fig. 5, the practical CFL, with which the spurious oscillations behind the front should vanish, has to be less than 2.6. This means that the present scheme allows a time integration interval that requires an Eulerian scheme to integrate at least three times.

4. Conclusions

We devised a high-order conservative semi-Lagrangian scheme for the fallout of raindrops. A cubic polynomial is used for the interpolation reconstruction. The positive definiteness of the scheme can be easily enforced through a slope modification. Although the scheme is free of the restriction for computational stability in terms of CFL number, the practical time step interval is actually bounded because of the variations in the velocity field. For the nonlinear model in the present paper, the practical CFL for the scheme should be less than 2.6. When compared with low-order schemes, the present method is much superior in capturing the finescale structures of disturbances.

The present scheme is in fact a high-order conservative scheme that allows a large time integration interval for a nonlinear rain sedimentation model. Thus, we believe that the scheme possesses the essential features for a numerical method of that sort. However, as mentioned in Pellerin et al. (1995) and Grabowski and Smolarkiewicz (1996), when coupled to condensation forcing terms, the scheme may need some extra modifications. Incorporating the present scheme into the real condensation model will be further explored.

As meteorological models include more and more direct computations of various physics, high-order schemes, such as the present one, will become increasingly necessary.

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REFERENCES

Bartello, P., and S. J. Thomas, 1996: The cost-effectiveness of semi-Lagrangian advection. *Mon. Wea. Rev.,* **124,** 2883–2897.

Bermejo, R., and J. Conde, 2002: A conservative quasi-monotone semi-Lagrangian scheme. *Mon. Wea. Rev.,* **130,** 423–430.

Colella, R., and P. R. Woodward, 1984: The piece-wise parabolic method (PPM) for gas-dynamical simulations. *J. Comput. Phys.,* **54,** 174–201.

Grabowski, W. W., and P. K. Smolarkiewicz, 1996: Two-time-level

semi-Lagrangian modeling of precipitating clouds. *Mon. Wea. Rev.,* **124,** 487–497.

Holnicki, P., 1995: A shape-preserving interpolation: Application to semi-Lagrangian advection. *Mon. Wea. Rev.,* **123,** 862–870.

Kato, T., 1995: A box-Lagrangian rain-drop scheme. *J. Meteor. Soc. Japan,* **73,** 241–245.

——, and K. Saito, 1995: Hydrostatic and non-hydrostatic simulation of moist convection: The applicability of hydrostatic approximation to high resolution model. *J. Meteor. Soc. Japan,* **73,** 59–77.

Kessler, E., 1969: *On the Distribution and Continuity of Water Substance in Atmospheric Circulations. Meteor. Monogr.,* No. 32, Amer. Meteor. Soc., 84 pp.

Klemp, J. B., and R. B. Wilhelmson, 1978: The simulation of three-dimensional convective storm dynamics. *J. Atmos. Sci.,* **35,** 1070–1096.

Lin, S., and R. B. Rood, 1996: Multidimensional flux-form semi-Lagrangian transportation schemes. *Mon. Wea. Rev.,* **124,** 2046–2070.

McDonald, A., 1984: Accuracy of multiply-upstream, semi-Lagrangian advection schemes. *Mon. Wea. Rev.,* **112,** 1267–1275.

——, 1987: Accuracy of multiply-upstream, semi-Lagrangian advection schemes II. *Mon. Wea. Rev.,* **115,** 1446–1450.

McGregor, J. L., 1993: Economical determination of departure points for semi-Lagrangian models. *Mon. Wea. Rev.,* **121,** 221–230.

Pellerin, P., R. Laprise, and I. Zawadzki, 1995: The performance of a semi-Lagrangian transport scheme for the advection–condensation problem. *Mon. Wea. Rev.,* **123,** 3318–3330.

Purnell, D. K., 1976: Solution of the advective equation by upstream interpolation with a cubic spline. *Mon. Wea. Rev.,* **104,** 42–48.

Rasch, P. J., and D. L. Williamson, 1990: On shape-preserving interpolation and semi-Lagrangian transport. *SIAM J. Sci. Stat. Comput.,* **11,** 656–686.

Riishøjgaad, L. P., S. E. Cohn, Y. Li, and R. Menard, 1998: The use of spline interpolation in semi-Lagrangian transport models. *Mon. Wea. Rev.,* **126,** 2008–2016.

Smolarkiewicz, P. K., and J. A. Pudykiewicz, 1992: A class of semi-Lagrangian approximations for fluids. *J. Atmos. Sci.,* **49,** 2082–2096.

Staniforth, A., and J. Côté, 1991: Semi-Lagrangian integration scheme for atmospheric model—A review. *Mon. Wea. Rev.,* **119,** 2206–2223.

Temperton, C., and A. Staniforth, 1987: An efficient two-time-level semi-Lagrangian semi-implicit integration scheme. *Quart. J. Roy. Meteor. Soc.,* **113,** 1025–1039.

van Leer, B., 1979: Toward the ultimate conservative difference scheme. V. A second order sequel to Godunov's method. *J. Comput. Phys.,* **32,** 101–136.

Williamson, D. L., J. G. Olson, and B. A. Boville, 1998: A comparison of semi-Lagrangian and Eulerian tropical climate simulations. *Mon. Wea. Rev.,* **126,** 1001–1012.

Xiao, F., 2000: A class of single-cell high-order semi-Lagrangian advection schemes. *Mon. Wea. Rev.,* **128,** 1165–1176.

——, and T. Yabe, 2001: Completely conservative and oscillationless semi-Lagrangian schemes for advection transportation. *J. Comput. Phys.,* **170,** 498–522.

Yabe, T., and T. Aoki, 1991: A universal solver for hyperbolic-equations by cubic-polynomial interpolation. 1. One-dimensional solver. *Comput. Phys. Commun.,* **66,** 219–232.

——, R. Tanaka, T. Nakamura, and F. Xiao, 2001: Exactly conservative semi-Lagrangian scheme (CIP-CSL) in one dimension. *Mon. Wea. Rev.,* **129,** 332–344.